

EM and the Forward–Backward Algorithm for Hidden Markov Models

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Abstract

These notes derive the forward-backward algorithm and EM (Baum-Welch) for hidden Markov models directly, by maximum likelihood. This is the machinery underneath every `.fit(method="em")` call in this afternoon’s coding tutorial — today we derive what that call is actually doing, down to the weighted linear regression at the heart of the tutorial’s AR-HMM M-step. Based on [stats305c/em](#) and [stats305c/hmms](#).

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1 Hidden Markov Models

A **hidden Markov model (HMM)** pairs a Markov chain of discrete latent states $z_1, \dots, z_T \in \{1, \dots, K\}$ with a sequence of observations $x_1, \dots, x_T$, one emitted per state. The joint distribution factors as

$$p(z_{1:T}, x_{1:T}; \Theta) = p(z_1; \pi_0) \prod_{t=2}^T p(z_t \mid z_{t-1}; P) \prod_{t=1}^T p(x_t \mid \theta_{z_t}), \quad (1)$$

with three parameter groups $\Theta = (\pi_0, P, \{\theta_k\}_{k=1}^K)$:

1. **Initial distribution:** $z_1 \sim \text{Cat}(\pi_0)$.
2. **Transition matrix:** $P \in [0, 1]^{K \times K}$ (row-stochastic), $P_{ij} = p(z_{t+1}=j \mid z_t=i)$.
3. **Emissions:** $x_t \sim p(\cdot \mid \theta_{z_t})$ — e.g. Gaussian, Poisson, or, for this afternoon’s tutorial, an *autoregressive* Gaussian, $x_t \mid z_t=k, x_{t-1} \sim \mathcal{N}(A_k x_{t-1} + b_k, Q_k)$.

We want the **maximum likelihood estimate** of Θ : the parameters that make the observed data $x_{1:T}$ most probable, marginalizing over the states we never observe,

$$\hat{\Theta} = \arg \max_{\Theta} \log p(x_{1:T}; \Theta), \quad p(x_{1:T}; \Theta) = \sum_{z_{1:T}} p(z_{1:T}, x_{1:T}; \Theta). \quad (2)$$

The sum in (2) is over all K^T possible state sequences — already intractable to evaluate for, say, $K = 6$ and a few hundred time steps, let alone to differentiate for gradient ascent. **EM** sidesteps this by exploiting the Markov structure of (1): it turns out we only ever need two low-dimensional summaries of the posterior over paths, not the whole K^T -entry table, and those summaries can be computed by the **forward-backward algorithm** in $O(TK^2)$ time. That’s the plan for the rest of these notes: derive EM in general, including the M-step for a simple Gaussian emission model (Section 2), then derive forward-backward as the way to compute the summaries EM’s E-step needs (Section 3). The M-step for the tutorial’s autoregressive emissions is left as an exercise at the end of Section 2.

2 The EM Algorithm

2.1 Jensen’s inequality and the ELBO

Lemma (Jensen’s inequality). *For any concave function f and random variable Y , $f(\mathbb{E}[Y]) \geq \mathbb{E}[f(Y)]$, with equality iff Y is constant (or f is linear). Since $\log$ is concave, $\log \mathbb{E}[g] \geq \mathbb{E}[\log g]$ for any positive random variable g .*

Introduce an **auxiliary distribution** $q(z_{1:T})$ over the whole hidden path — for now, completely arbitrary, with the same support as the true posterior. Multiply and divide by q inside the sum in (2), then apply Jensen’s inequality:

$$\begin{aligned} \log p(x_{1:T}; \Theta) &= \log \sum_{z_{1:T}} p(z_{1:T}, x_{1:T}; \Theta) \\ &= \log \sum_{z_{1:T}} q(z_{1:T}) \frac{p(z_{1:T}, x_{1:T}; \Theta)}{q(z_{1:T})} \\ &= \log \mathbb{E}_q \left[\frac{p(z_{1:T}, x_{1:T}; \Theta)}{q(z_{1:T})} \right] \\ &\geq \mathbb{E}_q \left[\log p(z_{1:T}, x_{1:T}; \Theta) - \log q(z_{1:T}) \right] \\ &\triangleq \mathcal{L}(\Theta, q). \end{aligned} \quad (3)$$

$\mathcal{L}(\Theta, q)$ is the **evidence lower bound (ELBO)**. It holds for *any* q , so we’ve replaced one hard problem, maximizing $\log p(x_{1:T}; \Theta)$ over Θ , with an easier joint problem: maximize $\mathcal{L}(\Theta, q)$ over (Θ, q) by coordinate ascent. That coordinate ascent is EM — E-step optimizes q , M-step optimizes Θ — and it is preferable to direct gradient ascent on (2) because, as we’ll see, each coordinate update has a simple, closed form.

2.2 The E-step

Note that $\log p(z_{1:T}, x_{1:T}; \Theta) = \log p(z_{1:T} \mid x_{1:T}; \Theta) + \log p(x_{1:T}; \Theta)$. Expanding the joint distribution inside the expectation in (3) and rearranging terms — a procedure we might call *ELBO surgery* — we obtain another form of the bound,

$$\mathcal{L}(\Theta, q) = \log p(x_{1:T}; \Theta) - D_{\text{KL}}(q(z_{1:T}) \parallel p(z_{1:T} \mid x_{1:T}; \Theta)), \quad (4)$$

where $D_{\text{KL}}(q||p) = \sum_z q(z) \log \frac{q(z)}{p(z)}$ is the **Kullback-Leibler (KL) divergence**. Since the KL divergence is non-negative and zero iff $q = p$, we see that $\mathcal{L} \leq \log p(x_{1:T}; \Theta)$ always, and maximizing $\mathcal{L}$ over q (with Θ fixed) means minimizing that KL divergence. The unique minimizer is

$$\boxed{q^*(z_{1:T}) = p(z_{1:T} \mid x_{1:T}; \Theta),} \quad (5)$$

the exact posterior over the whole path, at which point the bound is tight: $\mathcal{L}(\Theta, q^*) = \log p(x_{1:T}; \Theta)$. This is the **E-step**.

The catch: q^* is a distribution over K^T paths — we can't just write down a table of that size, let alone use it directly in the M-step below. Fortunately, as we will see, the M-step only needs *posterior marginals*, and we can compute those efficiently thanks to the Markovian nature of the model.

2.3 The M-step

The M-step maximizes the ELBO with respect to the parameters Θ , holding the posterior q^* fixed. Let's focus on the emission parameters $\{\theta_k\}_{k=1}^K$. As a function of those parameters, the ELBO is

$$\mathcal{L}(\Theta, q) = \mathbb{E}_q [\log p(x_{1:T}, z_{1:T}; \Theta)] + c \quad (6)$$

$$= \sum_{t=1}^T \sum_k \underbrace{q(z_t = k)}_{\triangleq \gamma_t(k)} \log p(x_t; \theta_k) + c, \quad (7)$$

where $\gamma_t(k)$ is the probability of being in state k at time t under the posterior q . Maximizing with respect to the parameters θ_k is simply a weighted maximum likelihood estimation problem. Often, the solution can be found in closed form.

Example: Gaussian emissions Let's consider a simple case where the emission model is Gaussian with mean θ_k ,

$$p(x_t; \theta_k) = \mathcal{N}(x_t \mid \theta_k, \sigma^2 I). \quad (8)$$

The ELBO as a function of θ_k is

$$\mathcal{L}(\Theta, q) = \sum_{t=1}^T \gamma_t(k) \log \mathcal{N}(x_t \mid \theta_k, \sigma^2 I) \quad (9)$$

$$= \sum_{t=1}^T \gamma_t(k) \left(-\frac{1}{2\sigma^2} (x_t - \theta_k)^\top (x_t - \theta_k) \right). \quad (10)$$

It's a little bit of calculus to show that this objective is maximized at

$$\hat{\theta}_k = \frac{\sum_{t=1}^T \gamma_t(k) x_t}{\sum_{t=1}^T \gamma_t(k)}. \quad (11)$$

Intuitively, we simply set the mean parameter equal to the average of observations, weighted by the probability of being in state k .

Exercise Now consider an autoregressive observation model (dropping the bias term for simplicity; see the remark below), $p(x_t | x_{t-1}; \theta_k) = \mathcal{N}(x_t | A_k x_{t-1}, Q_k)$. Show that the optimal dynamics parameters are obtained by the **weighted normal equations**, with solution

$$\hat{A}_k = \left(\sum_{t=1}^T \gamma_t(k) x_t x_{t-1}^\top \right) \left(\sum_{t=1}^T \gamma_t(k) x_{t-1} x_{t-1}^\top \right)^{-1}. \quad (12)$$

This is exactly **weighted least squares**: regress x_t on x_{t-1} , weighting time step t by how confident the E-step is that $z_t = k$. (The tutorial's actual fit includes a bias term too, $A_k x_{t-1} + b_k$; recover it with the usual trick for affine regression — augment x_{t-1} with a constant 1 and fold b_k into A_k as an extra column.)

2.4 EM as a minorize-maximize algorithm

Each iteration:

1. **E-step** (minorize) — build $\mathcal{L}(\Theta^{(i)}, q)$, which touches $\log p(x_{1:T}; \Theta)$ at $\Theta = \Theta^{(i)}$;
2. **M-step** (maximize) — set $\Theta^{(i+1)} = \arg \max_{\Theta} \mathcal{L}(\Theta, q^{(i)})$.

Since the M-step can only increase $\mathcal{L}$, and $\mathcal{L} \leq \log p(x_{1:T}; \Theta)$ always, the marginal log-likelihood is **guaranteed to increase monotonically** every iteration, with no step size to tune. EM converges to a *local* optimum, and (like K -means) is sensitive to initialization; random restarts are standard practice.

3 The Forward-Backward Algorithm

Our target quantity is the marginal $\gamma_t(k) = p(z_t=k | x_{1:T})$ needed for the M-step above. The key to computing it efficiently is the following factorization.

Claim:

$$p(z_t=k | x_{1:T}) \propto \underbrace{p(z_t=k | x_{1:t-1})}_{\triangleq \alpha_t(k)} p(x_t | z_t=k) \underbrace{p(x_{t+1:T} | z_t=k)}_{\triangleq \beta_t(k)}. \quad (13)$$

Proof, by two applications of Bayes' rule. Split $x_{1:T} = (x_{1:t-1}, x_t, x_{t+1:T})$ and apply Bayes' rule twice, each time treating one more block of observations as newly-arrived evidence. Conditioning on $x_{t+1:T}$ last,

$$p(z_t=k | x_{1:t-1}, x_t, x_{t+1:T}) \propto p(x_{t+1:T} | z_t=k, x_{1:t-1}, x_t) p(z_t=k | x_{1:t-1}, x_t).$$

In an HMM, once we know z_t , everything that happens afterward is independent of everything that happened before — the defining Markov property — so $p(x_{t+1:T} | z_t=k, x_{1:t-1}, x_t) = p(x_{t+1:T} | z_t=k) = \beta_t(k)$. Applying Bayes' rule once more to the remaining factor, now splitting off x_t ,

$$p(z_t=k | x_{1:t-1}, x_t) \propto p(x_t | z_t=k, x_{1:t-1}) p(z_t=k | x_{1:t-1}).$$

Since x_t depends only on z_t , $p(x_t | z_t=k, x_{1:t-1}) = p(x_t | z_t=k)$, and $p(z_t=k | x_{1:t-1}) = \alpha_t(k)$ by definition. Multiplying the pieces back together gives (13). $\square$

So the whole problem reduces to computing two quantities recursively: the **forward message** $\alpha_t(k) \triangleq p(z_t=k | x_{1:t-1})$ and the **backward message** $\beta_t(k) \triangleq p(x_{t+1:T} | z_t=k)$.

3.1 The forward algorithm

Claim: $\alpha_{t+1}(j) = \frac{1}{C_t} \sum_{k=1}^K P_{kj} \alpha_t(k) p(x_t | z_t=k)$, where $C_t \triangleq p(x_t | x_{1:t-1}) = \sum_k \alpha_t(k) p(x_t | z_t=k)$.

Proof, by induction on t . Base case: $\alpha_1(k) = p(z_1=k | x_{1:0}) = p(z_1=k) = \pi_{0,k}$ — there is no data to condition on before $t = 1$.

Inductive step: suppose $\alpha_t(k)$ is known for every k . First, incorporate the new observation x_t via Bayes' rule to get the *filtered* distribution,

$$p(z_t=k | x_{1:t}) = \frac{p(x_t | z_t=k) \alpha_t(k)}{\sum_{k'} p(x_t | z_t=k') \alpha_t(k')} = \frac{p(x_t | z_t=k) \alpha_t(k)}{C_t}.$$

Then predict one step ahead: by the law of total probability and the Markov property (z_{t+1} depends on the past only through z_t),

$$\alpha_{t+1}(j) = p(z_{t+1}=j | x_{1:t}) = \sum_k p(z_{t+1}=j | z_t=k) p(z_t=k | x_{1:t}) = \sum_k P_{kj} \frac{\alpha_t(k) p(x_t | z_t=k)}{C_t}. \quad \square$$

Writing $\ell_t = [p(x_t | z_t=1), \dots, p(x_t | z_t=K)]^\top$ for the vector of emission likelihoods at time t , this is the matrix-vector recursion

$$\alpha_{t+1} = \frac{P^\top (\alpha_t \odot \ell_t)}{C_t}, \quad C_t = \alpha_t^\top \ell_t, \quad (14)$$

costing $O(K^2)$ per step ($O(TK^2)$ total, versus $O(K^T)$ for naive enumeration). Notice normalization wasn't an afterthought bolted on for numerical stability — it fell straight out of the derivation, because α_t is a genuine conditional distribution (Bayes' rule) rather than an unnormalized joint probability. As a bonus, $C_t = p(x_t | x_{1:t-1})$ is exactly the one-step-ahead predictive likelihood, so by the chain rule we get the **marginal log-likelihood for free**:

$$\log p(x_{1:T}) = \sum_{t=1}^T \log C_t. \quad (15)$$

This is the quantity we track as “the log-likelihood” during EM.

3.2 The backward algorithm

Claim: $\beta_t(k) = \sum_{j=1}^K P_{kj} p(x_{t+1} | z_{t+1}=j) \beta_{t+1}(j)$.

Proof, by induction on t , downward from T . Base case: $\beta_T(k) = p(x_{T+1:T} | z_T=k) = 1$, since the “future after T ” is the empty sequence of observations, which is certain.

Inductive step: suppose $\beta_{t+1}(j)$ is known for every j . Introduce z_{t+1} by marginalizing,

$$\beta_t(k) = p(x_{t+1}, x_{t+2:T} | z_t=k) = \sum_j p(z_{t+1}=j | z_t=k) p(x_{t+1} | z_{t+1}=j, z_t=k) p(x_{t+2:T} | z_{t+1}=j, z_t=k, x_{t+1}).$$

By the Markov property, both x_{t+1} and $x_{t+2:T}$ depend on z_t only through z_{t+1} , so this reduces to $\sum_j P_{kj} p(x_{t+1} | z_{t+1}=j) \beta_{t+1}(j)$. $\square$

In matrix form, $\beta_t = P(\beta_{t+1} \odot \ell_{t+1})$. The backward pass runs from $t = T$ down to $t = 1$, the mirror image of the forward pass.

A note on normalization. Unlike α_t , β_t is not a normalized distribution over k — it’s a product of $T - t$ likelihood terms, so it can just as easily underflow for large T . But we’re free to rescale it: since $\gamma_t(k) \propto \alpha_t(k) \ell_{t,k} \beta_t(k)$ only needs $\beta_t(k)$ up to an overall constant, one that doesn’t depend on k , we can renormalize $\beta_t(\cdot)$ by any convenient constant B_t at every step of the recursion — e.g. $\beta_t(k) \leftarrow \beta_t(k)/B_t$ with $B_t = \sum_k \beta_t(k)$ — without changing $\gamma_t(k)$ once it’s normalized over k . This is the same numerical trick as for α_t , just applied by hand here instead of falling out of the derivation automatically.

3.3 Putting it together

One forward pass and one backward pass give $\alpha_t(k)$ and $\beta_t(k)$ for every t , and hence, via (13), the posterior marginal

$$\gamma_t(k) = p(z_t=k \mid x_{1:T}) \propto \alpha_t(k) \ell_{t,k} \beta_t(k). \quad (16)$$

This is the **E-step**, computed in $O(TK^2)$ time, without ever writing down the K^T -entry posterior over paths.

Exercise The transition-matrix M-step needs one more quantity: the pairwise marginal $\xi_t(i, j) \triangleq p(z_t=i, z_{t+1}=j \mid x_{1:T})$. Using the same conditional-independence argument as the proof of (13) — but now keeping *both* z_t and z_{t+1} fixed, rather than marginalizing the transition between them — show that

$$\xi_t(i, j) \propto \alpha_t(i) \ell_{t,i} P_{ij} \ell_{t+1,j} \beta_{t+1}(j)$$

(normalize over i, j to get an honest probability). This is exactly the weight $\hat{P}_{ij} \propto \sum_{t=1}^{T-1} \xi_t(i, j)$ needs in the transition-matrix M-step.

References

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